

The Space Needle Through the Toolkit

The BB(6) cryptid that machine 3 was built to mirror — halting on an exact power of 2

1. Overview

The Space Needle is a Busy Beaver BB(6) cryptid (machine 1RB1LA_1LC0RE_1LF1LD_0RB0LA_1RC1RE_---0LD, discovered by mx dys, January 2025). It is the **multiplicative-coincidence** archetype of the bbchallenge catalogue — and the machine our own machine 3 was constructed to mirror, with powers of 2 in place of powers of 3. This note runs the collection's toolkit over it, both to characterize it and to check that the tools built on our machines reproduce the community's findings on a genuine catalogued cryptid. They do, exactly.

Results: the one-variable form reproduces the wiki's published trajectory exactly; the machine provably cannot cycle (the state is its own potential); halting is equivalent to the orbit hitting an exact power of 2 (Theorem 1); the branch statistic is the 2-adic valuation law; and the measured per-step log-growth is 0.6515, matching the wiki's 0.652355 (so $b \approx 1.918^n$ vs the wiki's 1.92006). It is a divergent cryptid, obviously non-halting; the open core is the same single-orbit question as everywhere.

2. The machine

The wiki gives two high-level forms. **Doucette's one-variable form** is the halting-equivalent reduction we analyze. Writing $v(b)$ for the 2-adic valuation of b and $m = b / 2^{v(b)}$ for its odd part, from $b_0 = 6$:

HALT if b is an exact power of 2 ($m = 1$); else $b \rightarrow b + v(b) + (3/2)(m - 1)$.

(The term $(3/2)(m-1)$ is always an integer because m is odd.) **Fidelity:** the implementation reproduces the wiki's sequence 6, 10, 17, 41, 101, 251, 626, 1095, 2736, 2995 exactly. **Ducharme's low-level form** $A(b, c) \rightarrow (1, c)$ halts, $(2b, c) \rightarrow (2+5b+c, 1)$, $(2b+1, c) \rightarrow (b-1, 3+b+c)$, start $(3, 1)$ — is also implemented and runs without halting, as a cross-check on the machine level.

3. Analysis

3.1 Theorem 1: halting is hitting a power of 2

The map halts precisely when b is an exact power of 2, so the halting question is: does the orbit of 6 ever land on $\{1, 2, 4, 8, 16, \dots\}$? This is the **multiplicative-coincidence** type — a geometric target of exact powers, spaced by a factor of 2. It is machine 3's halting criterion ($a = 3^j$) with base 2 in place of base 3, and without machine 3's exponent congruence: here every power of 2 halts.

3.2 Theorem 2: no cycles (the state is its own potential)

For any non-halting b the odd part $m \geq 3$, so $b \rightarrow b + v(b) + (3/2)(m-1)$ increases b by at least 3. Thus b is strictly increasing and the machine cannot cycle: it halts or escapes to infinity. This is simpler than machine 3, whose a was not monotone and needed the potential $\Phi = a + b$; here the single state

variable serves directly, exactly as for the Hydra family. (Verified: minimum increment 3 over all non-power-of-2 $b < 300,000$, and strict increase along 200,000 orbit steps.)

3.3 Branch statistic and growth

The branch is $v(b)$, the 2-adic valuation. Measured over 200,000 steps it matches the geometric law $P(v = k) = 2^{-(k+1)}$ (0.498, 0.252, 0.125, 0.063, ... vs 0.5, 0.25, 0.125, 0.0625) — the same valuation law behind machine 3 (base 3) and the Collatz shortcut. From it the growth constant follows: when b is odd (half the time) $b \rightarrow \sim 5b/2$, and averaging over the valuation law the per-step log-growth is

$$\text{mean } \Delta \log b = 0.6515 \text{ (measured) vs } 0.652355 \text{ (wiki), so } b \approx 1.918^n.$$

Powers of 2 are spaced by $\times 2$ while b grows by about $\times 1.92$ per step, so the orbit passes roughly one power-of-2 scale per step; the chance of landing *exactly* on one decays like $1/b$, a convergent sum. So Space Needle is a **divergent-type** cryptid, probvously non-halting — the same shape as machine 3, and the reason machine 3 was a faithful analogue to build.

4. Status and verification

check	scope	result
one-variable form vs wiki	reproduces 6, 10, 17, 41, 101, 251, 626, 1095, 2736, 2995 exactly	pass
Theorem 1 (halt = power of 2)	by construction of the map	pass
Theorem 2 (no cycles)	min increment 3 for $b < 300,000$; strict increase over 200,000 orbit steps	pass
branch statistic	$v(b)$ matches $2^{-(k+1)}$ to three decimals over 200,000 steps	pass
growth constant	0.6515 measured vs 0.652355 (wiki); b reached 187,989 bits in 200,000 steps, no power of 2 hit	pass

Artifacts: needle.py (both forms + wiki fidelity), theorems.py (T1–T4 and measurements). This report is deliberately kept separate from the collection meta report for now. **Status: unchanged — open (divergent, multiplicative-coincidence cryptid); the toolkit reproduces the community’s findings and adds the no-cycle proof beneath them.**